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Treatment of Locally Advanced Prostate Cancer

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Keywords

Advanced prostate cancer; androgen deprivation; high-risk prostate cancer; radiation therapy; radical prostatectomy; tumor staging

Questions

1. Identification of patients with high-risk prostate cancer is best achieved by:
    a. transrectal ultrasonography.
    b. serum prostate-specific antigen (PSA).
    c. digital rectal examination.
    d. serum PSA, biopsy grade, clinical stage.
    e. PSA kinetics.
2. By using the Kattan postoperative nomogram, which of the following contributes most to the risk of biochemical recurrence after radical prostatectomy?
    a. Positive surgical margin
    b. Pretreatment serum PSA of 17 ng/mL
    c. Gleason 4 + 3 disease
    d. Established capsular penetration
    e. Seminal vesicle invasion

3. Neoadjuvant androgen deprivation (AD) before radical prostatectomy leads to:
    a. improved biochemical-free survival.
    b. improved overall survival.
    c. reduced positive surgical margins.
    d. reduced local recurrence.
    e. increased operative morbidity.
4. In men with locally advanced prostate cancer undergoing prostatectomy, clinical overstaging (i.e., pathologically organ confined disease) occurs in:
    a. less than 10%.
    b. 15% to 30%.
    c. 40% to 60%.
    d. 70% to 80%.
    e. more than 90%.
5. The use of high-dose antiandrogen monotherapy after prostatectomy in men with locally advanced disease:
    a. reduces disease progression.
    b. increases cardiac morbidity.
    c. does not have an impact on sexual function.
    d. improves overall survival.
    e. improves local disease control.
6. In men with locally advanced/high-risk prostate cancer, the most effective treatment among the following options is:
    a. brachytherapy + external-beam radiation therapy.
    b. neoadjuvant AD + external-beam radiation therapy.
    c. neoadjuvant AD + external-beam radiation therapy + adjuvant AD.
    d. concurrent AD plus external-beam radiation therapy.
    e. long-term AD alone.
7. Risk assessment schemes for prostate cancer are most accurate for patients with:
    a. low-risk disease.
    b. high-risk disease.
    c. the disease.
    d. metastatic disease.

e. locally advanced cancers.

8. The current appropriate dose for adjuvant radiation therapy after radical prostatectomy is:
   a. less than 45 Gy.
   b. 45 to 50 Gy.
   c. 51 to 55 Gy.
   d. 56 to 60 Gy.
   e. greater than 60 Gy.
9. The use of AD in combination with radiation therapy for those with high-risk cancers is associated with all of the following EXCEPT:
   a. improved local control.
   b. improved biochemical-free survival.
   c. less gastrointestinal toxicity.
   d. worsened sexual function.
   e. more urinary frequency.
10. The benefits of early radiation therapy after radical prostatectomy in men with locally advanced disease are observed:
   a. for improved local control.
   b. for improved overall survival.
   c. in men with positive surgical margins.
   d. none of the above.
   e. all of the above.
11. After radical prostatectomy in those with loco-regional disease, overall survival may be optimized by adjuvant treatment with:
   a. androgen deprivation.
   b. radiation therapy.
   c. sipuleucel-T.
   d. docetaxel.
   e. radiation therapy + androgen deprivation.
12. Over the past decade, the trend in the treatment of men with higher-risk disease (CAPRA 6-10) has shown:
   a. increase in primary androgen deprivation.
   b. decrease in primary androgen deprivation.
   c. increase in radiation therapy.
   d. decrease in radical prostatectomy.
   e. decrease in active surveillance/watchful waiting.



FIG. 159.1 Computed Tomography Scan of Pelvis.

Imaging

1. A 65-year-old man underwent a radical retropubic prostatectomy 4 years ago for a pT3a N0 Gleason 8 adenocarcinoma of the prostate. Two years postoperatively, his PSA first became detectable and has slowly risen since then to its current value of 1.2 ng/mL. A computed tomography (CT) scan of the pelvis is obtained and is depicted in Fig. 159.1. The most likely diagnosis is:
    a. retained seminal vesicle.
    b. enlarged lymph node.
    c. recurrence in the prostatectomy bed.
    d. rectal mass.
    e. rectal diverticulum.

Answers

1. **d. Serum PSA, biopsy grade, clinical stage.** Although clinical stage, serum PSA, and Gleason score all individually predict pathologic stage and prognosis, the combination of these three variables increases the accuracy of this assessment.
2. **b. Pretreatment serum PSA of 17 ng/mL. Despite the trend toward lower serum PSA at the time of diagnosis, PSA remains an important predictor of treatment failure, and greater elevations (>8 ng/mL) of PSA contribute significantly to calculated biochemical recurrence.**
3. **c. Reduced positive surgical margins. The randomized and nonrandomized studies of neoadjuvant androgen deprivation in men with lower clinical stage (cT1-T2) clearly demonstrate a reduction in the rate of positive surgical margins; however, this advantage has not been observed in men with cT3c and has not translated into improved long-term PSA-free survival.**
4. **b. 15% to 30%.** Recent data suggest that clinical overstaging occurs in approximately 27% of men with clinical stage T3 disease undergoing prostatectomy, consistent with the range in the literature of 7% to 26%.
5. **a. Reduces disease progression.** Bicalutamide at greater dose (150 mg) appears to have a positive effect in those men with locally advanced disease, with 43% reduction in disease progression and potential benefit of improved survival; however, it should be remembered that high-dose bicalutamide given to men with localized prostate cancer is associated with increased risk of death (hazard ratio: 1.23).
6. **c. Neoadjuvant AD + external-beam radiation therapy + adjuvant AD. The accumulated data from multiple Radiation Therapy Oncology Group (RTOG) and European Organisation for Research and Treatment of Cancer (EORTC) trials suggest that improved outcomes are achieved with greater duration of administration of androgen deprivation in combination with external-beam radiation therapy, with apparent benefit of both neoadjuvant and adjuvant therapy.**
7. **a. Low-risk disease.** Validation has confirmed the general accuracy of the available risk assessment tools, but there is a tendency to

overestimate the risk of cancer recurrence in men with high-risk disease features.

8. **e. Greater than 60 Gy.** There is a trend to improve response to adjuvant radiation therapy, and most contemporary series report doses greater than 60 Gy, with potential threshold of either 61.2 or 64 Gy. Similarly, for primary radiation therapy, improved outcomes have been shown for higher doses (78 Gy or greater).
9. **c. Less gastrointestinal toxicity.** The longer application (longer than 6 to 9 months) of AD in conjunction with radiation therapy may be associated with increased rectal morbidity as well as sexual dysfunction.
10. **e. All of the above. Data from EORTC 22911 and Southwest Oncology Group (SWOG) 8794 clearly demonstrate a benefit of adjuvant radiation therapy in men with pT3 disease, after radical prostatectomy, with respect to biochemical relapse-free, metastasis-free, and overall survival, as well as improved local control.** The EORTC study suggests that patients who benefit the most are those with positive surgical margins.
11. **e. Radiation therapy + androgen deprivation. The traditional approach to those patients with N+ disease after radical prostatectomy has been androgen deprivation therapy, based on results from ECOG 3886 showing improved overall survival for immediate versus delayed treatment.** More recent studies suggest that the combination of androgen deprivation and radiotherapy may yield the best outcomes with respect to overall survival, even when compared with androgen deprivation alone. There is no evidence that adjuvant immunotherapy or chemotherapy improves outcomes in these patients after surgery.
12. **b. Decrease in primary androgen deprivation. There has been significant evolution in the management of high-risk prostate cancer.** A contemporary analysis of the CaPSURE database has shown a significant increase in the number of men undergoing radical prostatectomy with concomitant decrease in primary androgen deprivation as well as radiation therapy. Rates of surveillance/watchful waiting appear stable.

Imaging

1. **c. Recurrence in the prostatectomy bed.** The pelvic CT scan demonstrates a mass in the prostatectomy bed on the right at the level of the urethra-vesicle anastomosis. Because the mass is anterior to the rectum, it is not likely to be a lymph node or seminal vesicle. In view of the radical prostatectomy specimen and the rising PSA, the mass is likely a prostate cancer recurrence in the prostatectomy bed.

Chapter Review

1. At least 10% of men with newly diagnosed prostate cancer have locally advanced disease.
2. Risk assessment for locally advanced disease is best determined by a combination of PSA, T stage, cancer grade, and extent of cancer in the biopsy.
3. PSA recurrence following radical prostatectomy is influenced by Gleason score, extracapsular extension, seminal vesicle invasion, positive lymph nodes, and positive surgical margins.
4. Neoadjuvant androgen deprivation therapy before radical prostatectomy has no role.
5. Early androgen deprivation therapy appears to have a potential survival advantage in subsets of men with more aggressive disease. Unfortunately, side effects of the therapy may be a sequela.
6. The role of adjuvant radiation therapy following radical prostatectomy is controversial. A subset of patients apparently benefits from adjuvant radiation therapy. Unfortunately, all studies to date are flawed such that specific subsets of patients who will benefit have not been adequately defined.
7. Patients with seminal vesicle involvement or regional lymph node metastases are highly likely to develop progressive disease despite adjuvant local therapy.
8. EORTC trials suggest that improved outcomes are achieved with greater duration of administration of androgen deprivation in

combination with external-beam radiation therapy for selected patients with high-grade disease.

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Management Strategies for Biochemical Recurrence of Prostate Cancer

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Keywords

biochemical recurrence; prostate cancer imaging; prostate cancer; salvage cryotherapy; salvage prostatectomy; salvage radiation

Questions

1. What have the American Urological Association (AUA) and European Association of Urology (EUA) determined as the definition of prostate-specific antigen (PSA) failure following radical prostatectomy?
    a. Any level of detectable PSA following radical prostatectomy.
    b. Two values of 0.1 ng/mL or higher.
    c. 0.2 ng/mL.
    d. 0.2 ng/mL with a confirmatory value.
    e. 0.4 ng/mL with a confirmatory value.
2. Which imaging modality in PSA recurrence following radical prostatectomy has the highest sensitivity at the lowest PSA values?
    a. Radionucleotide bone scan.
    b. Computed tomography (CT) of abdomen/pelvis.

c. Fluorodeoxyglucose-positron emission tomography (FDG-PET) scan.
d. Prostascint scan.
e. Multiparametric magnetic resonance imaging (MRI).

3. What is the minimum recommended dosage for salvage radiotherapy?
   a. 43 Gy
   b. 54 Gy
   c. 64 Gy
   d. 72 Gy
   e. 78 Gy
4. Candidates for salvage prostatectomy following failed radiation therapy of the prostate should have all but which factor before treatment?
   a. Negative metastatic workup.
   b. At least 10 years of life expectancy.
   c. Biopsy-proven local recurrence.
   d. Negative pelvic lymph node sampling.
   e. PSA value less than 10 ng/mL.
5. Which has the highest sensitivity in detecting recurrent local disease following radiation therapy for prostate cancer?
   a. Rectal exam.
   b. Transrectal ultrasound.
   c. MRI.
   d. Prostascint scan.
   e. PET scan.
6. Which of these is NOT a commonly known side effect of androgen deprivation therapy (ADT)?
   a. Decreased bone mineral density.
   b. Mania.
   c. Hot flashes.
   d. Fatigue.
   e. Sexual side effects.
7. The most important therapeutic consideration in selecting either local salvage therapy or systemic therapy for a patient with a rising PSA value after definitive local therapy is:

a. patients with a rising PSA level after definitive local therapy should be started on hormonal therapy because they are destined to experience systemic relapse.
b. patients with a rising PSA level should undergo salvage local procedures, such as radiation or cryotherapy or prostatectomy, before undergoing any systemic treatment.
c. patients with a rising PSA level and no metastatic disease should be started on chemotherapy.
d. patients with a rising PSA level should undergo neither systemic nor local treatments, because the only appropriate context in which to begin any intervention is when radiographic metastases have developed.
e. patients with a rising PSA level should be risk stratified and treated with a modality of therapy that matches their risk of relapse, risk of developing local versus systemic disease, and risk of dying of other causes.

Answers

1. **d. 0.2 ng/mL with a confirmatory value.** Throughout the radical prostatectomy literature, there have been more than 50 individual definitions for PSA failure. In 2007, the AUA Guidelines panel for localized prostate cancer released its recommendations for PSA failure following radical prostatectomy as 0.2 ng/mL with a confirmatory value greater than 0.2 ng/mL (Cookson et al., 2007). [a] Although higher values would result in greater specificity for disease recurrence and progression, the value of 0.2 ng/mL resulted in higher sensitivity and generalizability. The panel reported that in no way should this individual definition be used for determining the usage of adjuvant/salvage therapies and reiterated that this definition is not predictive of death outcomes.
2. **e. Multiparametric magnetic resonance imaging (MRI).** Traditional imaging techniques such as CT scan and bone scan demonstrate limited value at PSA values less than10 ng/mL (Dotan et al., 2005; Okotie et al., 2004). Multiparametric MRI has shown

reliability in identifying local recurrence even at low levels of PSA. In fact, MRI sensitivity may be as high as 86% at PSA values between 0.4 and 1.4 ng/mL and has also demonstrated an ability to perform better than PET-CT (Panebianco et al., 2012; Sciarra et al., 2008).

3. **c. 64 Gy.** In 1999, the American Society for Therapeutic Radiation Oncology (ASTRO) guidelines panel concluded that a minimum of 64 Gy should be used for salvage radiation following radical prostatectomy (Cox JD et al., 1999). This dosage was confirmed by the AUA guidelines in 2013 (Thompson IM et al., 2013). **Modern reports suggest that salvage radiation dosages as high as 76 Gy may demonstrate effective biochemical recurrence-free survival with reasonable toxicities** (De Meerleer et al., 2008; Ost, Lumen et al., 2011). It is hoped that studies such as SAKK 09/10 will clarify the role of dose escalation and identify an optimal therapeutic window for salvage radiation (**http://clinicaltrials.gov/show/NCT01272050**).
4. **d. Negative pelvic lymph node sampling.** Although modern series of salvage prostatectomy demonstrate improved morbidity, patient selection is of the utmost importance (Heidenreich et al., 2010; Stephenson, Scardino et al., 2004). **Patients who are candidates for salvage surgery must have at least 10 years of life expectancy, a negative metastatic workup, biopsy-proven local recurrence, and ideally a PSA value less than 10 ng/mL.** A separate negative pelvic lymph node sampling is not imperative before undergoing salvage surgery, although sending lymph nodes for frozen section before prostatectomy would not be unreasonable.
5. **c. MRI.** Traditional imaging modalities have not demonstrated consistent ability to detect radiorecurrent prostate cancer. Transrectal ultrasonography (TRUS) demonstrates sensitivities no better than digital rectal exam, and CT scans are not relevant at PSA values less than 20 ng/mL (Crook J et al., 1993). Alternatively, MRI with contrast enhancement, spectroscopy, and diffusion-weighted imaging has demonstrated improved sensitivity (Haider et al., 2008; Hara et al., 2012; Westphalen et al., 2010). In fact, diffusion-weighted imaging has demonstrated a sensitivity and

specificity of 100% using a 22-core biopsy as a reference (Hara et al., 2012).

6. **b. Mania.** All other answers are commonly reported side effects of patients undergoing ADT.
7. **e. Patients with a rising PSA level should be risk stratified and treated with a modality of therapy that matches their risk of relapse, risk of developing local versus systemic disease, and risk of dying of other causes.** In the setting of biochemical recurrence following radical prostatectomy, patients' risks of metastasis and death are variable (Pound et al., 1999). **Both metastasis-free and prostate-cancer survival depend on several factors such as Gleason score, PSA doubling time, and time from surgery to biochemical recurrence** (Antonarakis et al., 2012; Freedland et al., 2005, 2006). All of these factors must be considered in salvage therapies. Clearly, patients with significant comorbidities and limited life expectancy are not good candidates for salvage local therapies, and this should be taken into consideration. The use of nomograms to help in decision-making processes is paramount.

Chapter Review

1. The AUA and the EAU have defined biochemical recurrence following radical prostatectomy as a value of ≥0.2 ng/mL with a second confirmatory value.
2. Salvage radiation therapy should be discussed with patients with non-metastatic prostate cancer and a rising PSA following radical prostatectomy.
3. Androgen deprivation therapy should be considered in patients receiving salvage radiation therapy, especially in those with high-risk features.
4. Multiparametric MRI can be a useful tool for detection of local recurrence following radical prostatectomy.
5. Patients who undergo a transient rise in their PSA should be closely monitored, because not all men are destined for biochemical failure,

let alone clinical failure.

6. The ideal candidates for salvage radical prostatectomy have at least 10 years of life expectancy, lack of metastasis on imaging, and a PSA less than 10 ng/mL.

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[a] Sources referenced can be found in *Campbell-Walsh Urology, 12th Edition,* on the Expert Consult website.

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Hormonal Therapy for Prostate Cancer

Scott Eggener

Keywords

androgen ablation; androgen deprivation; androgen receptor blockade; locally advanced prostate cancer; metastatic prostate cancer

Questions

1. The effectiveness of estrogen as a hormone therapy for prostate cancer is primarily based on:
    a. direct cytotoxic effects of estrogen on prostate cancer cells.
    b. competitive binding of estrogen to the androgen receptor.
    c. inhibition of the conversion of cholesterol to pregnenolone.
    d. desensitizing luteinizing hormone–releasing hormone (LHRH) receptors in the anterior pituitary.
    e. negative feedback on luteinizing hormone (LH) secretion by the pituitary.
2. The expected response of a man to the administration of the nonsteroidal antiandrogens is:
    a. LH increases, testosterone decreases, estrogen decreases.
    b. LH increases, testosterone increases, estrogen decreases.
    c. LH increases, testosterone increases, estrogen increases.
    d. LH decreases, testosterone decreases, estrogen increases.

e. LH decreases, testosterone increases, estrogen increases.

3. All of the following therapeutic approaches for androgen axis blockade are in current clinical use, EXCEPT:
    a. inhibition of androgen synthesis.
    b. blocking androgen action by binding to the androgen receptor in a competitive fashion.
    c. ablating the source of androgens.
    d. direct inhibition of androgen receptor–mediated pathways.
    e. inhibition of LHRH and LH release.
4. Nonsteroidal antiandrogens:
    a. do not act as agonists for prostate cancer cells when used in combination with LHRH agonists.
    b. allow long-term maintenance of erectile function and sexual activity at rates similar to men undergoing surgical castration.
    c. commonly induce gastrointestinal toxicity, manifest as constipation leading, on occasion, to fecal impaction.
    d. have pancreatic toxicity, ranging from reversible mild to fulminant, life-threatening suppurative pancreatitis, and require periodic monitoring of serum amylase and lipase.
    e. cause fluid retention and thromboembolism in the majority of patients.
5. Which of the following nonsteroidal antiandrogens is associated with interstitial pneumonitis and a delayed adaptation to darkness after exposure to bright illumination?
    a. Bicalutamide
    b. Flutamide
    c. Hydroxyflutamide
    d. Nilutamide
    e. Cyproterone acetate
6. Which of the following statements is TRUE concerning LHRH agonists?
    a. Based on a review of 24 trials, involving more than 6600 patients, survival after therapy with an LHRH agonist was significantly better than after surgical castration.
    b. Although depot preparations and osmotic pump devices allow dosing to extend from 28 days to 1 year, the most effective

dosing regimen is daily.

c. Current LHRH agonists are based on analogues of the native LHRH decapeptide by amino acid substitutions, particularly position 6 of the peptide.

d. Widespread use of orally effective LHRH agonists has been limited by severe allergic reactions in some patients, even after previously uneventful treatment.

e. The use of LHRH agonists is limited to combined androgen blockade.

7. Each of the following has been associated with a favorable initial response to androgen deprivation therapy (ADT), EXCEPT:
   a. the magnitude of the PSA decline.
   b. the rapidity of the PSA decline.
   c. the PSA doubling time before initiating ADT.
   d. the Gleason score of the primary tumor.
   e. the maintenance of a detectable PSA.
8. Which of the following statements about the complications of ADT is TRUE?
   a. Most men undergoing ADT have normal bone mineral density before initiating therapy, and it usually takes at least a decade of treatment before the average man will develop osteopenia.
   b. Hot flashes occur in about one quarter of men on ADT but should always be treated because of the associated rare but life-threatening cardiovascular side effects.
   c. Erectile dysfunction after surgical castration or use of an LHRH is common but not inevitable: Although 1 in 5 men maintains some sexual activity, only 1 in 20 maintains high levels of sexual interest (libido).
   d. Because most men on ADT maintain lean muscle mass, the increase in weight is due to increases in adipose tissue.
   e. Gynecomastia and mastodynia are common with estrogenic compounds and antiandrogens but are effectively treated by external beam radiation after they occur.
9. Which of the following statements about the combination of 3 months of neoadjuvant ADT before radical prostatectomy is TRUE?

a. Positive surgical margin rates are significantly reduced with ADT-treated patients.
b. There is a significant reduction in biochemical (PSA) progression with ADT-treated patients.
c. The benefit of neoadjuvant ADT appears to be in men with locally advanced disease and/or those with high-grade disease.
d. Antiandrogen monotherapy has not shown a significant reduction of biochemical failure, but LHRH agonists have demonstrated this reduction.
e. Although the results of prospective randomized studies of this combination are mixed, the overall body of evidence supports the use of ADT in this setting.

10. Combined androgen blockade:
    a. is designed to address the low levels of testicular androgens remaining after the use of LHRH agonists or antagonists.
    b. typically uses an antiandrogen at the time of PSA rise after treatment with an LHRH agonist.
    c. has not shown a survival advantage compared with an LHRH agonist alone.
    d. significantly benefits men with minimally metastatic disease when used in combination with surgical castration.
    e. using cyproterone acetate has a slightly worse outcome.
11. Compared with deferred ADT, early ADT instituted before the development of objective metastatic disease:
    a. provides an overall survival advantage in all clinical disease states.
    b. has an equivalent quality of life.
    c. does not increase overall death rates.
    d. does not prevent the emergence of castration-resistant prostate cancer.
    e. should be offered to men with PSA recurrence after radical prostatectomy because of the rapid disease progression in this clinical setting.
12. In men with lymph-node metastatic prostate cancer discovered at the time of radical prostatectomy:

a. significant overall survival benefits of immediate ADT are limited to those with extrapelvic positive nodes.
b. a significant overall survival benefit of immediate ADT has been demonstrated in men who have also undergone subsequent radical prostatectomy.
c. a significant overall survival benefit of immediate ADT has been demonstrated in men who have not undergone subsequent radical prostatectomy.
d. b and c are correct.
e. a, b, and c are correct.

13. From a strictly financial point of view, which of the following forms of ADT is the least expensive?
    a. Scrotal orchiectomy
    b. LHRH agonist
    c. Diethylstilbestrol (DES)
    d. Antiandrogen monotherapy
    e. LHRH antagonist
14. In a trial studying continuous ADT to intermittent ADT for men with rising PSA after radiation therapy:
    a. prostate cancer–specific death was more common in the continuous ADT arm.
    b. death unrelated to prostate cancer was more common in the intermittent ADT arm.
    c. attrition from the intermittent arm was rare throughout the cycles.
    d. duration of intermittent ADT progressively shortened over time.
    e. quality of life was not improved in the intermittent arm.
15. There is general consensus that ADT should always be initiated in a hormonally intact patient in which of the following clinical settings?
    a. Before radical prostatectomy with a clinical T2 tumor.
    b. In all clinical stages of men undergoing external beam radiation therapy.
    c. In those with clinically localized prostate cancer who do not want local treatment.

d. In those with symptomatic metastatic disease.
e. In those with high-grade prostatic intraepithelial neoplasia (PIN) on needle biopsy who refuse a subsequent biopsy.

Answers

1. **e. Negative feedback on luteinizing hormone (LH) secretion by the pituitary.** After the success of surgical castration in treating prostate cancer, the first central inhibition of the hypothalamic-pituitary-gonadal axis exploited the potent negative feedback of estrogen on LH secretion. **Estradiol is a 1000-fold more potent at suppressing LH and follicle-stimulating hormone (FSH) secretion by the pituitary compared with testosterone.** Although estrogen has some direct cytotoxic effects on prostate cancer cells, this is not its primary mode of action. **All antiandrogens competitively bind to the androgen receptor. Aminoglutethimide inhibits the conversion of cholesterol to pregnenolone, an early step in steroidogenesis. The LHRH agonists desensitize LHRH receptors in the anterior pituitary.**
2. **c. LH increases, testosterone increases, estrogen increases. Unlike the steroidal antiandrogens, such as cyproterone acetate, which have central progestational inhibitory effects, the nonsteroidal antiandrogens simply block androgen receptors, including those in the hypothalamic-pituitary axis.** Because those central androgen receptors no longer sense the normal negative feedback exerted by testosterone, both LH levels and—as the normal testicular response to increased LH—testosterone levels increase. **Peripheral conversion of this excessive testosterone also increases estrogen levels, leading to the gynecomastia and mastodynia associated with the nonsteroidal antiandrogens.**
3. **d. Direct inhibition of androgen receptor–mediated pathways.** There are four therapeutic approaches for androgen axis blockade in current clinical use. All current forms of ADT function by reducing the ability of androgen to activate the androgen receptor,

whether through lowering levels of androgen or by blocking androgen–androgen receptor binding. **Therefore, the androgen receptor is not directly affected by ADT, leading many to hypothesize that hormone-refractory prostate cancer is a reactivation of androgen receptor–mediated pathways.**

4. **b. Allow long-term maintenance of erectile function and sexual activity at rates similar to men undergoing surgical castration.** By blocking testosterone feedback centrally, the nonsteroidal antiandrogens cause LH and testosterone levels to increase, allowing antiandrogen activity without inducing hypogonadism and potency can be preserved. In clinical trials specifically examining erectile function and sexual activity in men on antiandrogen monotherapy, however, long-term preservation of those domains was 20% and not significantly different from that in men undergoing surgical castration. **All antiandrogens can act agonistically on prostate cancer cells and, when used in combination with LHRH agonists, withdrawal of the antiandrogen can lead to declines in PSA and even objective responses.** The common gastrointestinal toxicity is diarrhea, most often seen with flutamide. Liver toxicity, ranging from reversible hepatitis to fulminant hepatic failure, is associated with all nonsteroidal antiandrogens and requires periodic monitoring of liver function tests. The steroidal antiandrogen cyproterone acetate is associated with fluid retention and thromboembolism.
5. **d. Nilutamide.** About one quarter of men on nilutamide therapy will note a delayed adaptation to darkness after exposure to bright illumination, and in approximately 1% of patients, nilutamide is also associated with interstitial pneumonitis, which can progress to pulmonary fibrosis. Hydroxyflutamide is the active metabolite of flutamide. Cyproterone acetate is a steroidal antiandrogen.
6. **d. Widespread use of orally effective LHRH agonists has been limited by severe allergic reactions in some patients, even after previously uneventful treatment. The LHRH agonists exploit the desensitization of LHRH receptors in the anterior pituitary following chronic exposure to LHRH, thereby shutting down the production of LH and, ultimately, testosterone.** Analogues of

native LHRH increase their potency and half-lives. **The initial flare in LH and testosterone may last 10 to 20 days, and co-administration of an antiandrogen is required for only 21 to 28 days.** Survival after therapy with an LHRH agonist was equivalent to that for orchiectomy. The clinical utility of the first LHRH agonists was hampered by their short half-lives, requiring daily dosing. The LHRH antagonist abarelix has been associated with severe allergic reactions: All LHRH agonists are administered either IM or subcutaneously. LHRH can be used without combination with an antiandrogen.

7. **e. The maintenance of a detectable PSA.** The odds ratio of progressing to androgen-refractory progression at 24 months after starting ADT was 15-fold higher in those who did not achieve an undetectable PSA. **The magnitude and rapidity of PSA decline, the pre-ADT PSA doubling time, and pretreatment testosterone levels are all associated with the ability to predict the response to ADT.** For each unit increase in Gleason score, the cumulative hazard of castration-resistant progression was nearly 70%.
8. **c. Erectile dysfunction after surgical castration or use of an LHRH is common but not inevitable: Although 1 in 5 men maintains some sexual activity, only 1 in 20 maintains high levels of sexual interest (libido).** The loss of sexual functioning is not inevitable with surgical or chemical castration, with up to 20% of men able to maintain some sexual activity. **Libido is more severely compromised, with approximately 5% maintaining a high level of sexual interest. More than half of men undergoing ADT meet the bone mineral density criteria for osteopenia or osteoporosis; it is estimated osteopenia will develop in the average man within 4 years of initiating ADT.** Hot flashes are among the most common side effects of ADT, affecting between 50% and 80% of patients. Hot flashes should be treated only in those who find them bothersome. **Loss of muscle mass and increase in percent fat body mass are common in men undergoing ADT. Prophylactic radiation therapy (10 Gy) has been used to prevent or reduce gynecomastia and mastodynia,**

but it has no benefit once these side effects have already occurred.

9. **a. Positive surgical margin rates are significantly reduced with ADT-treated patients.** In both nonrandomized and randomized clinical trials, the pathologic positive surgical margin rate is significantly reduced. In one study, the positive surgical margin rate fell from nearly 50% in hormonally intact patients to 15% in ADT-treated patients. **Despite this improvement, there has not been a corresponding significant reduction in biochemical (PSA) progression in ADT-treatment patients, a finding in four separate prospective randomized studies. The benefit of ADT in men with locally advanced disease and/or high-grade disease has been in combination with external beam radiation therapy.** There is no evidence that any form of 3-month neoadjuvant ADT before radical prostatectomy reduces biochemical failure rates.
10. **e. Using cyproterone acetate has a slightly worse outcome.** In studies of combined androgen blockade using the steroidal antiandrogen cyproterone acetate compared to LHRH agonists alone, the outcomes were slightly worse with the combination, suggesting increased non–prostate cancer deaths with this agent. Combined androgen blockade is designed to block the possible contribution of adrenal androgens to prostate cancer progression. Combined androgen blockade uses an antiandrogen along with an LHRH agonist: Addition of an antiandrogen at the time of PSA rise (evidence of hormone-refractory disease) is considered secondary hormonal manipulation. Several clinical trials have shown a slight but significant survival advantage for combined androgen blockade. **A landmark randomized clinical trial comparing surgical castration alone with surgical castration combined with flutamide did not show a significant benefit in men with minimal metastatic disease.**
11. **d. Does not prevent the emergence of castration-resistant prostate cancer.** The timing of the initiation of ADT has not prevented the development of castration-resistant prostate cancer. **Although early ADT may provide an overall survival advantage in certain clinical disease states, in most studies**

there is no significant overall survival advantage. Indeed, in localized, low-risk prostate cancer, early ADT is associated with an increase in overall death rates. The natural history of disease progression after biochemical failure following radical prostatectomy is protracted: Median time to bone metastases is 10 years.

12. **b. A significant overall survival benefit of immediate ADT has been demonstrated in men who have also undergone subsequent radical prostatectomy.** A randomized prospective study of men with positive regional pelvic lymph nodes discovered at the time of radical prostatectomy showed an overall survival advantage to immediate ADT. In that study, all men also underwent the radical prostatectomy. A similar study, performed by the European Organisation for Research and Treatment of Cancer (EORTC) in men who did not undergo radical prostatectomy if positive nodes were discovered did not show a significant survival advantage to immediate ADT.
13. **c. Diethylstilbestrol (DES).** At a dose of 1 to 3 mg/day with no prophylactic breast irradiation, DES is the cheapest form of ADT. LHRH agonists would be cheaper than scrotal orchiectomy only if the patient lived a few months after the administration of ADT. Combined androgen blockade is the most expensive form of ADT.
14. **d. Duration of intermittent ADT progressively shortened over time.** The duration of intermittent ADT progressively shortened over time: the median interval of the "off cycle" was 20.1 months for the first interval, 13.2 months for the second cycle, 9.1 months for the third, and 4 to 5 months thereafter. Disease-specific death (prostate cancer and related treatments) was more common in the intermittent-therapy arm compared to the continuous-therapy, 120 versus 94, respectively; conversely, deaths unrelated to prostate cancer were more common in the continuous-therapy arm compared with the intermittent-therapy arm, 162 versus 148, respectively. Attrition from intermittent androgen deprivation progressively increased over time as patients either developed castration-resistant prostate cancer or died of another cause. Attrition occurred in only 5% of men in the first interval, whereas

68% had stopped intermittent therapy by the third interval. A secondary end point, improved quality of life in the intermittent-therapy arm, was associated with significantly better scores for hot flashes, desire for sexual activity, and urinary symptoms.

15. **d. In those with symptomatic metastatic disease.** In hormonally intact men with symptomatic metastatic prostate cancer, ADT is always indicated. There is no significant biochemical (PSA) disease-free advantage in men treated with neoadjuvant ADT. **The benefits of ADT in combination with external beam radiation therapy are in men with locally advanced and/or high-grade disease.** The use of ADT in men with low-risk, localized prostate cancer is associated with a significantly lower overall survival. There is no indication for ADT in the management of PIN.

Chapter Review

1. Antiandrogens bind to the androgen receptor in a competitive fashion. They are either steroidal or nonsteroidal.
2. Steroidal antiandrogens suppress LH release. Thus the steroidal antiandrogens block the effects of testosterone on the receptor as well as lower testosterone through their progestational central inhibition effect. The nonsteroidal antiandrogens simply block androgen receptors.
3. When performing an orchiectomy, double-ligating the transected segments of the cord with one being a transfixion suture is advised.
4. Initial exposure to LHRH agonists results in a flare of testosterone that may last for up to 20 days; the clinical effects of the flare may be blocked by the simultaneous administration of an antiandrogen. The two drugs should be administered together, and the antiandrogen therapy should be continued for 3 weeks.
5. After 4 years of androgen deprivation, the average man is osteopenic.
6. Bothersome hot flashes can be treated with megestrol, cyproterone, DES, estradiol, medroxyprogesterone, or gabapentin.
7. ADT may result in bone loss, sexual dysfunction, hot flashes, decreased cognitive function, loss of muscle mass, increase in body

fat, anemia, gynecomastia, diabetes, and increased cardiovascular mortality.

8. Twenty percent of individuals with metastatic prostate cancer die of nonprostate cancer causes.
9. Bilateral orchiectomy reduces testosterone by 90% within 24 hours.
10. The androgen receptor remains responsive to androgen even in the castration-resistant state; therefore, ADT should be continued in the patient who has castration-resistant prostate cancer.
11. The steroid synthesis inhibitors aminoglutethimide, ketoconazole, and abiraterone require simultaneous glucocorticoid replacement. Aminoglutethimide requires mineral corticoid replacement as well, whereas abiraterone results in increased mineral corticoid production that may result in hypokalemia and hypertension.
12. Intermittent ADT compared with continuous therapy is not superior and may be worse.
13. Many men with prostate cancer will never require ADT because of the protracted course of the disease.
14. Estradiol is a 1000-fold more potent at suppressing LH and FSH secretion by the pituitary compared to testosterone.
15. Therapies that block the androgen receptor but do not lower testosterone, such as nonsteroidal antiandrogens, may produce excessive amounts of testosterone, which is peripherally converted to estrogens, leading to the gynecomastia and mastodynia.
16. All antiandrogens can act agonistically on prostate cancer cells and, when used in combination with LHRH agonists, withdrawal of the antiandrogen can lead to declines in PSA and even objective responses.
17. The LHRH agonists exploit the desensitization of LHRH receptors in the anterior pituitary following chronic exposure to LHRH, thereby shutting down the production of LH and, ultimately, testosterone.
18. Prophylactic radiation therapy (10 Gy) has been used to prevent or reduce gynecomastia and mastodynia, but it has no benefit once these side effects have already occurred. Once they are present, a simple mastectomy may be required in some men for cosmetic and symptomatic relief.

19. A clinical trial comparing surgical castration alone with surgical castration combined with flutamide did not show a significant benefit in men with minimal metastatic disease.
20. Although early ADT may provide an overall survival advantage in certain clinical disease states, in most studies there is no significant overall survival advantage. Indeed, in localized, low-risk prostate cancer, early ADT is associated with an increase in overall death rates.
21. The benefits of ADT in combination with external beam radiation therapy are limited to men with locally advanced and/or high-grade disease.

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Treatment of Castration-Resistant Prostate Cancer

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Keywords

androgen receptor; castration-resistant prostate cancer; cytotoxic chemotherapy; immunotherapy; bone health

Questions

1. The term “castration-resistant disease” has replaced the older classification of “hormone-resistant disease” used to define all patients who demonstrate evidence of disease progression after initial androgen deprivation treatment because:
    a. androgen deprivation treatment focusing on medical/surgical castration is the initial systemic approach for patients with metastatic prostate cancer.
    b. in patients with castration-resistant disease, responses to subsequent androgen receptor (AR)-targeted treatments continue to show a benefit.
    c. current data indicate that the AR continues to play a major role in the control of prostate cancer growth even when serum levels of testosterone are in the castrate range (<50 ng/dL).
    d. castration resistance does not equal hormone resistance.
    e. **all of the above adequately describe the castration-resistant state.**

2. Patients eventually stop benefiting from primary and secondary hormonal treatments and become refractory. Which of the following statement(s) INCORRECTLY defines the “castration-resistant” paradigm?
    a. Careful clinical monitoring of patients in clinical practice, including regular physical exams and sequential assessments of radiologic parameters and serum prostate-specific antigen (PSA) levels, facilitates early identification of patients who are becoming resistant to secondary hormonal manipulations.
    b. The definition of castration resistance is based on well-defined clinical and pathologic criteria such as Gleason score and extent of disease.
    c. Determination of castration resistance requires clinical evidence of PSA or disease progression to primary and secondary hormonal manipulations.
    d. The vast majority of patients treated with endocrine manipulations will develop evidence of disease progression and eventually require chemotherapy.
3. Docetaxel remains the standard first-line chemotherapy treatment for patients with metastatic castration-resistant prostate cancer considered candidates for this modality. Which of the following statement(s) is NOT true?
    a. Docetaxel infusion every 3 weeks (for as many as 10 cycles) given in conjunction with daily oral prednisone is the standard schedule that has been shown to prolong survival and improve quality of life compared with mitoxantrone.
    b. Significant clinical benefits are seen in patients regardless of their age, functional status, and presence or absence of pain.
    c. Patients with metastatic castration-resistant disease who have favorable functional status (fully ambulatory, asymptomatic), no visceral involvement, and normal hemoglobin and serum lactate dehydrogenase (LDH) have survival outcomes using docetaxel that frequently exceed 2 years.
    d. Frequent toxicities associated with docetaxel treatment include fatigue, myelosuppression, neuropathy, lacrimation, and nail

changes, among others. Routine clinical and laboratory evaluation before each cycle is indicated.

e. Patients demonstrating rising serum PSAs during the first three to four cycles of treatment should probably be taken off treatment because it is not effective.

4. Cabazitaxel is another taxane approved for metastatic castration-resistant prostate cancer. In preclinical studies that used cancer cell lines and mouse xenograft models, cabazitaxel was shown to be active in docetaxel-sensitive tumors, as well as those with primary or acquired docetaxel resistance. Which of the following statements adequately describes the clinical experience with this compound?
   a. Cabazitaxel was shown to prolong survival compared with mitoxantrone in patients previously treated with docetaxel.
   b. The toxicity pattern of cabazitaxel suggests a lower incidence of neuropathy, fatigue, lacrimation, and nail changes but has a higher incidence of neutropenic fever and diarrhea compared with docetaxel.
   c. Data on the TROPIC trial suggest that patients who fail to respond or develop early evidence of disease progression after docetaxel may benefit from cabazitaxel.
   d. Patients demonstrating disease progression after docetaxel can still survive longer than 12 months with cabazitaxel treatment.
   e. All of the above are correct.
5. After initial gonadal suppression, AR signaling is upregulated in castration-resistant disease and continues to play a major role in tumor growth. Which of the following statement(s) adequately describes AR-targeted treatments?
   a. CYP17 inhibitors target androgen synthesis both of adrenal and intracrine source and yield significant benefit in patients treated with first-line gonadal suppression who subsequently develop evidence of disease progression. This effect has been shown in patients with or without previous treatment with docetaxel.
   b. The side effects with abiraterone acetate include a mineralocorticoid excess state (efficiently prevented by a concomitant administration of prednisone), fatigue, abnormal

liver function tests, possible cardiac toxicity, and potential drug interactions.

c. Enzalutamide is a nonsteroidal antiandrogen that differs from the first-generation compounds (flutamide, bicalutamide, and nilutamide) based on a greater AR affinity, AR nuclear translocation, and DNA binding. Most patients with castration-resistant disease benefit from treatment, which was shown to be significantly superior to placebo in prospective randomized trials.

d. The benefits from CYP17 inhibitors (abiraterone) and AR antagonists (enzalutamide) are most likely more pronounced in patients who have not received prior docetaxel treatment.

e. All of the above are correct.

6. Which of the following statements is correct regarding patients with widely metastatic prostate cancer who present with back pain?

a. Administration of narcotic analgesics is the appropriate management, and if pain management becomes more challenging, patients should be referred to hospice.

b. All patients with known bone metastasis should be carefully assessed clinically for the possibility of epidural cord or nerve root compression. Administration of high-dose dexamethasone and early magnetic resonance imaging (MRI) should be used, and more definitive treatment with radiation or neurosurgical decompression should be considered.

c. Patients with back pain and stable skeletal radiographs should be treated with narcotic analgesics and corticosteroids. If improvement occurs, no further evaluation is necessary.

d. If the PSA is not rising and the workup with a bone scan and computed tomography (CT) scans reveal stable disease, it can be assumed that cord compression or other complex neurologic involvement is unlikely.

e. Most current systemic treatments are effective for managing extensive bone metastasis even if there is evidence of nerve root or cord compression.

7. A small proportion of patients with advanced-stage disease develop a syndrome of rapid and dramatic development of severe symptoms

(pain, obstruction, weight loss) and clinical evidence of rapidly growing disease with soft tissue and organ involvement. In these patients, typically serum PSA levels are either below detectable levels or grossly disproportionate to the other extent of disease parameters. Which statement is correct regarding this rare event in castration-resistant prostate cancer?

a. These patients usually benefit from hormonal therapy including all AR-targeted compounds.
b. Serum PSA is low or undetectable, and the PSA stains in tumor biopsies are usually negative. These tumors do not express AR.
c. A small proportion of patients demonstrate evidence of a rapidly progressing disease predominantly involving visceral sites, with low or no PSA expression. Biopsy is indicated because it will affect treatment decisions.
d. Patients with this clinical syndrome often demonstrate anaplastic tumors at biopsy, some with small-cell features, and most stain positive for neuroendocrine markers. Platinum-based chemotherapy has been shown to offer some benefits for patients with the small-cell variety, whereas those with anaplastic tumors that are not of the small-cell subtype should be treated with taxane-based chemotherapy.
e. All statements are correct except a.

8. The radiopharmaceutical radium-223 was approved for the treatment of patients with metastatic castration-resistant disease. Which statement regarding 223Ra is FALSE?

a. Radium-223 is an α-emitting radiopharmaceutical that has recently shown to be associated with a survival advantage compared to symptomatic/palliative care.
b. α particles are approximately 7000 times heavier than β particles, and as few as one or two hits can be sufficient to cause cell death, in comparison with hundreds or thousands of hits required from β particles. In addition, α particles have a very short path length (<100 μm), which may spare surrounding healthy bone marrow, thereby limiting hematologic toxicities.

c. Radium-223 infusion has shown a very favorable toxicity spectrum with low hematologic toxicity rates.
d. It is indicated for patients with bone metastasis, hemoglobin greater than 10 g/L, and lymph-node metastasis smaller than 3 cm.
e. It was approved only for patients who have received prior docetaxel treatment.

9. Sipuleucel-T is a personalized vaccine derived from autologous CD54 + dendritic cells, the major class of antigen-presenting cells, which are apheresed from individuals and processed with a recombinant fusion protein made up of prostatic acid phosphatase (PAP) and granulocyte-macrophage–colony-stimulating factor (GM-CSF). Which statement is TRUE regarding this treatment?
    a. Sipuleucel-T is approved for all patients with castration-resistant disease as long as they are symptomatic.
    b. Sipuleucel-T treatment results in PSA declines and prolongation of progression-free survival but no survival improvements.
    c. Sipuleucel-T should be offered to patients with no evidence of metastasis as long as their disease is castration resistant.
    d. Sipuleucel-T is a treatment option for patients with minimally or asymptomatic metastatic prostate cancer. Treatment is generally very safe. There is no evidence that sipuleucel-T treatment causes symptomatic relief, any clinically meaningful PSA declines, or delay in disease progression. The drug was approved based on a survival benefit compared with placebo.
10. The nonmetastatic castrate-resistant prostate cancer state, also known as M0 CRPC, is an evolving disease entity with its own therapeutic considerations. Which of these statements about the nonmetastatic CRPC state is FALSE?
    a. This clinical state is defined by a rising PSA level, castrate testosterone levels (<50 ng/dL), and a lack of radiographic metastases on conventional CT/MRI and nuclear medicine bone scans.
    b. Apalutamide was the first drug to be U.S. Food and Drug Administration (FDA) approved specifically for men with

nonmetastatic (M0) CRPC.

c. Enzalutamide is FDA approved for all patients with CRPC, including both those with metastatic (M1) CRPC and nonmetastatic (M0) CRPC.
d. Abiraterone acetate has also recently received FDA approval for use in men with M0 CRPC, as well as its prior approval for M1 mCRPC.
e. All of these statements are true, and none are false.

Answers

1. **e. All of the above adequately describe the castration-resistant state.**
2. **b. The definition of castration resistance is based on well-defined clinical and pathologic criteria such as Gleason score and extent of disease.**
3. **e. Patients demonstrating rising serum PSAs during the first three cycles of treatment should be taken off treatment because it is not effective.**
4. **e. All of the above are correct.**
5. **e. All of the above are correct.**
6. **b. All patients with known bone metastasis should be carefully assessed clinically for the possibility of epidural cord or nerve root compression. Administration of high-dose dexamethasone and early magnetic resonance imaging (MRI) should be used, and more definitive treatment with radiation or neurosurgical decompression should be considered.**
7. **e. All statements are correct except a.**
8. **e. It was approved only for patients who have received prior docetaxel treatment.**
9. **d. Sipuleucel-T is a treatment option for patients with minimally or asymptomatic metastatic prostate cancer. Treatment is generally very safe. There is no evidence that sipuleucel-T treatment causes symptomatic relief, any clinically meaningful**

PSA declines, or delay in disease progression. The drug was approved based on a survival benefit compared with placebo.

10. **d. Abiraterone acetate has also recently received FDA approval for use in men with M0 CRPC, as well as its prior approval for M1 mCRPC. The only agents with regulatory approval for M0 CRPC are apalutamide and enzalutamide.**

Chapter Review

1. Androgen ablation induces apoptosis.
2. The AR may be stimulated by hormones other than androgens, including estrogens and progestins, as well as growth factors and cytokines.
3. Patients with castration-resistant disease are not androgen independent and should be maintained on ablative hormonal therapy.
4. PSA doubling time (PSADT) may be used to predict bone scan progression and survival; a PSADT of less than 3 months is associated with a rapid clinical course.
5. When evaluating therapeutic agents, progression-free survival is a better end point than response rate.
6. PSA may or may not be affected by drugs that are efficacious, and therefore it is not a good marker to evaluate new drugs—perhaps circulating tumor cells will become a better marker.
7. Docetaxel is the first-line chemotherapeutic agent for metastatic castration-resistant prostate cancer.
8. Abiraterone inhibits enzymes involved in androgen synthesis. It does result in secondary mineralocorticoid excess with resultant hypertension and hypokalemia, and as such it is commonly given with prednisone. Occasionally, when secondary mineralocorticoid excess causes significant abnormalities, a mineralocorticoid antagonist may be necessary.
9. Bone metastases in prostate cancer are usually blastic; hypercalcemia is rare.

10. Suspected spinal cord compression from prostate metastases may be diagnosed with a spinal MRI. Those with compression or impending compression, if they are not androgen suppressed, should have an immediate orchiectomy, or be given aminoglutethimide or ketoconazole and high-dose corticosteroids. A decompression laminectomy and radiation therapy should be considered.
11. Bisphosphonates or denosumab is used to limit skeletal events. Oral calcium supplements and vitamin D may be necessary.
12. Radiopharmaceuticals used to treat bone pain due to prostate metastases include the β-emitters strontium-89 and samarium-153. These agents may cause severe myelotoxicity. The α-emitter radium-223 shows promise in palliating bone pain and improving survival without the myelosuppressive effects of the β-emitters.
13. Rarely, patients with advanced prostate cancer may have a transformation of their tumor to a neuroendocrine/anaplastic variant. These tumors are endocrine resistant, frequently involve the viscera and brain, have little impact on PSA, and are treated with platinum-based chemotherapy.
14. After initial gonadal suppression, AR signaling is upregulated in castration-resistant disease and continues to play a major role in tumor growth.
15. Sipuleucel-T is a treatment option for patients with minimally or asymptomatic metastatic prostate cancer. Treatment is generally very safe. There is no evidence that sipuleucel-T treatment causes symptomatic relief, any clinically meaningful PSA declines, or delay in disease progression. The drug was approved based on a survival benefit compared with placebo.
16. Apalutamide and enzalutamide, but not abiraterone, are specifically approved by the FDA for use in men with M0 CRPC. Enzalutamide, but not apalutamide, is also FDA approved for M1 CRPC. Abiraterone is only approved for M1 CRPC.